Solution Design: Citizen-Led Wetland Monitoring Platform

Version: 0.1 — Draft **Date:** 2026-05-15 **Authors:** Joy Ghosh **Status:** Draft

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1. Background & Context

This document describes the solution design for the data platform built under the assignment "Technical Support to Implement Citizen-Led Data Generation and Management Activities", commissioned by the Nile Basin Discourse (NBD).

NBD brings together civil society, academic institutions, and government bodies across the Nile Basin to promote equitable, sustainable management of shared water resources. Wetland degradation — driven by encroachment, pollution, and climate variability — is a priority concern. Decision-makers lack regular, ground-level data on these wetlands. Without it, enforcement windows close before degradation is documented.

This assignment addresses that gap. It establishes a citizen-led monitoring system at two transboundary pilot sites:

- **Mara Basin** — shared between Kenya and Tanzania;
- **Sio Basin** — shared between Kenya and Uganda;

This document describes the technical backbone of that system. The platform collects data submitted by community members, processes and validates it, and presents results to partners and decision-makers through a public-facing portal.

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2. Problem Statement

Wetland managers and government authorities in the Nile Basin lack reliable, regular data on the health of transboundary wetlands. Formal monitoring is infrequent, expensive, and conducted by specialists who are rarely present on the ground.

Communities alongside the Mara and Sio-Siteko wetlands observe changes daily: unusual water colour, reduced fish catch, encroachment by farms, shifting flood patterns. This knowledge has no formal channel into decision-making systems.

This platform creates that channel. The design must respect three real-world constraints:

Connectivity. Many sampling sites have no mobile data coverage. The USSD channel requires only GSM voice coverage. KoboCollect stores data offline and syncs when connectivity is available.

Device access. Not all community members have smartphones. The pollution reporting workflow must be accessible from a basic feature phone.

Digital literacy. Wetland watchers and citizen scientists are not technical users. Every interaction must be guided, menu-driven, and completable without reading instructions.

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3. Solution Overview

Akvo will build a mobile-first, citizen-centric data platform. It enables community-driven data collection, administration, and visualisation through familiar and accessible tools.

The platform is an MVP — a foundational system supporting the two pilot basins. It can be extended if NBD decides to add new basins, wetlands, or data layers during scale-up.

The platform has three layers: data collection, administration, and visualisation. Two purpose-built workflows feed into a single public-facing wetland data portal.

Guiding Principles

Open source first. All custom components are built under open-source licences. KoboToolbox and all backend components are fully open source. Google Earth Engine, used for satellite data processing, has a generous free tier and a long-standing commitment to support non-commercial use.

Low-connectivity resilience. USSD runs on the GSM voice network only. KoboCollect captures data offline and syncs on reconnect.

Community data ownership. Data collected by citizens belongs to the communities and partner organisations. The platform is designed to facilitate handover to NBD without vendor lock-in.

Interoperability. Every monitoring site has a persistent, structured identifier (e.g. `NBD-MARA-001`). This identifier is embedded in the KoboCollect form, every admin layer record, every GEE pipeline output, every lab QA record, every FGD session record, and every portal API response. Any external dataset can join to platform data using this key alone — no custom field mapping is needed. Spatial layers are published as GeoJSON via documented API endpoints.

Scope

- Pollution episode reporting via USSD and WhatsApp (Kenya)
- Monthly structured water quality sampling via KoboToolbox (Tanzania — Mara wetlands; Kenya and Uganda — Sio-Siteko)
- Admin layer for data cleaning and approval
- Wetland data portal with pollution incident map, health scores, and management actions
- Sentinel 1 & 2 / CHIRPS Earth Observation data integration via Google Earth Engine
- Laboratory QA data ingestion
- FGD session data capture for indigenous knowledge

3.1 Constraints & Hardware Requirements

User-side Devices and Roles

Role	Description	Minimum device	Connectivity needed
Wetland watcher	Community member reporting pollution episodes (Kenya)	Basic feature phone	GSM only (USSD) or smartphone with WhatsApp
Citizen scientist	Trained volunteer conducting monthly structured water quality sampling (Tanzania, Kenya, Uganda)	Android smartphone with KoboCollect	Intermittent data (offline-capable)
CSO staff	UWASNET / KEWASNET staff socialising the platform and entering FGD session data	Laptop or tablet	Reliable medium to low bandwidth internet
Academic partner	Makerere / UoN staff conducting shadow sampling and lab QA	Any browser-capable device	Reliable internet
NBD / NDF / County / District officials	Strategic and operational data users accessing the wetland data portal	Any browser-capable device	Reliable internet

KoboCollect stores survey data on the device when there is no connectivity. It syncs automatically to the KoboToolbox server when the device reconnects.

Citizen scientists and wetland watchers are trained through a Train-the-Trainers (ToT) model. The platform must be fully usable from the first day of deployment by users who received training second-hand, without requiring IT support.

Handheld multi-parameter probes — used by citizen scientists to measure pH, temperature, and dissolved oxygen — are provided by the project. These are separate hardware items. The data they produce is entered through the KoboCollect form.

Hosting Infrastructure

- Hosting model: **Cloud-based, Dockerised**
- Production server: **8 GB RAM, 4-core CPU, 30 GB storage**
- Media attachments (photos, documents) stored in cloud object storage, separate from the VM

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4. Architecture Design

4.1 System Layers

The platform has three layers. The two data collection workflows maintain separate pipelines through the admin layer, then converge at the output stage.

Data Collection Layer — the citizen-facing interfaces. Wetland watchers submit pollution episodes via USSD or WhatsApp Business bot. Citizen scientists submit monthly structured sampling data via KoboCollect.

Admin / Processing Layer — the internal layer where staff clean and approve submissions. The pollution and sampling pipelines remain separate. Scoring is automated from approved data. External data (Sentinel 1 & 2, CHIRPS, lab QA results) and FGD session records are also ingested here.

Wetland Data Portal Layer — the public-facing output. The two pipelines converge in a single portal showing pollution incidents on an interactive map, wetland health scores, management actions, and satellite overlays. All views are accessible on mobile and can be exported as PDFs.

4.2 Pollution Episode Reporting (Kenya)

Actors: wetland watchers — community members who report pollution events as they occur.

Incident types reported: unusual water colour, smell, fish or other animal kills, changes in water levels.

Two parallel channels serve the same reporting flow:

Channel A — USSD (Feature Phone Users)

Feature phone users dial a dedicated short code and navigate a branching menu.

```
Wetland watcher dials short code
         │
         ▼
USSD branching menu presented
         │
         ├── Select incident type
         │            │
         ├── Select location
         │
         ▼
Report submitted to admin layer
```

The flow runs entirely over the GSM voice network. No mobile data, smartphone, or app installation required.

Channel B — WhatsApp Business Bot (Smartphone Users)

```
Wetland watcher opens WhatsApp
         │
         ▼
Menu-driven bot flow initiated
         │
         ├── Select incident type
         │
```

```
       │
       ├── Select sub-county
       │
       └── Attach GPS-stamped photo or voice note (optional)
       │
       ▼
Report submitted to admin layer
```

WhatsApp users can attach photos and voice notes alongside their submission. Both channels use Africa's Talking as the telco gateway and feed into the same admin-layer pipeline.

Admin Layer — Pollution Pipeline

1. **Ingest** — report received from USSD or WhatsApp webhook; written to the admin layer
2. **Clean** — staff review for completeness; remove obviously erroneous entries
3. **Approve** — submission approved for publication to the wetland data portal
4. **Traffic light classification** — approved records contribute to basin health status (Green / Yellow / Red)

4.3 Monthly Structured Water Quality Sampling (Tanzania, Kenya, Uganda)

Actors: citizen scientists — trained community volunteers who visit designated sampling sites once per month.

Collection tool: KoboToolbox, accessed via the KoboCollect Android app. KoboToolbox is open source and has robust offline functionality.

All numeric fields in the KoboCollect form have min-max constraints configured at the form level. This catches obvious probe entry errors before the form is saved. Out-of-range entries prompt the citizen scientist to re-enter the value before the form can be submitted.

Data Collection Flow

```
Citizen scientist arrives at sampling site (e.g. NBD-MARA-001)
       │
       ▼
Opens KoboCollect on Android smartphone (offline mode)
       │
       ▼
Completes geo-referenced structured form:
  ├── Physico-chemical measurements
  │
  ├── Ecological observations
  │
  ├── Hydrological observations
  │
  └── GPS-stamped photographs
       │
```

```
        ▼
Form saved locally on device (no connectivity required)
        |
        ▼
Connectivity available → KoboCollect syncs to KoboToolbox server
        |
        ▼
Submission received by KoboToolbox         |
        ▼
KoboToolbox API → processing service pulls new submissions
        |
        ▼
Admin layer — sampling pipeline
```

Admin Layer — Sampling Pipeline

Same steps as the pollution pipeline: Ingest → Clean → Approve. After approval, the scoring engine computes parameter group scores and the composite score automatically.

Shadow Sampling and Lab QA Integration

Every quarter, academic partners (Makerere University / University of Nairobi) conduct shadow sampling at the same sites as citizen scientists. They collect independent samples and test for all citizen scientist parameters, plus Biochemical Oxygen Demand, Orthophosphate, Nitrate, and Mercury.

Shadow sampling results are compared against citizen scientist submissions to validate measurements and flag systematic discrepancies for retraining.

Lab QA results are ingested as a separate record type in the admin layer, linked to the site identifier and sampling period.

4.4 Admin Interface

The admin interface is the internal processing layer. It is a web application accessible to authorised staff only,

Roles

Two roles are defined. Admin is a strict superset of Reviewer.

Role	Permissions
Reviewer	Review and approve or reject pollution reports, sampling records, lab QA reports, and FGD session records

Role	Permissions
Admin	All Reviewer permissions plus: invite users and assign roles; create and manage monitoring sites; delete records

Sidebar Navigation

Both roles see **Data** as the primary workspace item. Admins additionally see **User management** and **Site management** under a Management section. Reviewer accounts see these items but they are locked.

Data Screen

The Data screen is the central working view. All submitted records appear in a single filterable list. Three selectors narrow the list:

- **Form** — Pollution report · Sampling data · Lab QA report · FGD session
- **Status** — Pending approval · Approved
- **Basin** — Mara Basin · Sio-Siteko

The record count updates live as selectors change. A **Clear** control resets all three selectors.

Each row represents one submission. A Form chip (colour-coded by type) and a Basin / Site chip identify the record at a glance. Clicking a row expands it inline to show the full submission detail. The row collapses on a second click.

Actions available in the expanded view:

Status	Reviewer actions	Admin additional actions
Pending approval	Approve · Reject · Edit	Delete
Approved	View	Edit · Delete

For sampling data records, the expanded view shows the automated score and health class (A–E). These values are read-only — they are computed from submitted measurements.

Add New

A blue **+ Add New** button sits right-aligned in the filter bar. Clicking it opens a modal to select a form type and basin, then launches the corresponding webform.

Four form types are available:

Pollution report — basin-level; no specific site required.

Sampling data — the modal additionally prompts for the specific monitoring site.

Lab QA report — entry point for academic partners entering lab results. Links to the relevant sampling record and site.

FGD session — entry point for CSO staff entering structured indigenous knowledge data from Monthly Baraza sessions. The form captures: - Date of Baraza - Basin and associated sampling site - Number of FGD participants - Structured responses per dimension (dropdown): - Fish abundance change: Same or increased / Slightly declined / Moderately declined / Severely declined - Water clarity change: Same or clearer / Somewhat worse / Much worse - Vegetation cover change: Same or more / Partially lost / Severely lost - Pollution events reported: None / Occasional / Frequent - Open notes (stored but not scored) - Facilitator name

FGD records link to a basin and sampling period. The scoring pipeline reads the most recent FGD record for each site and period to compute the indigenous knowledge signal used in the fuzzy logic adjustment (see Section 5.6).

```
Monthly Baraza (FGD session)
          │
CSO staff enters FGD session record via Add New
          │
          ▼
Admin layer — FGD record
  site:           NBD-MARA-001
  period:         June 2026
  fish_decline:   moderately_declined  → 0.6
  clarity:        much_worse           → 1.0
  vegetation:     partially_lost       → 0.4
          │
          ▼
Scoring pipeline reads record
→ IK signal = average(0.6, 1.0, 0.4) = 0.67
→ fuzzy adjustment applied to composite score
```

Data added via Add New webforms is moved directly to Approved status.

User Management (Admin only)

Displays all platform users as cards showing name, email, organisation, and assigned role. Admins can invite new users by email and assign Admin or Reviewer roles. Users cannot self-register.

Authentication: the admin interface uses SSO via common identity providers (Google, Microsoft). No passwords are managed by the platform.

Site Management (Admin only)

Lists all registered monitoring sites in a table with columns for Site ID, name, country, basin, and coordinates. Admins can add new sites, edit site metadata, or disable a site (which hides it from the Add New form but retains its historical records). Site identifiers follow the format `NBD-MARA-001`.

Each site record supports attached documents. Supported formats: PDF, JPG, PNG (maximum 10 MB per file). Documents are stored in cloud object storage, linked to the site by Site ID. Attachments are visible to Admins and Reviewers in the expanded site view. They are not publicly visible on the portal.

Examples of attached documents: wetland management plans, baseline assessment reports, field notes, boundary maps.

4.5 Wetland Data Portal

The portal gives NBD Secretariat, NDFs, county and district officials, CSOs, academic partners, and communities access to collected data, derived health scores, and management actions.

Key features: - **Pollution incident map** — geospatial display of reported pollution episodes by basin - **Wetland health scores** per site, per time period - Trend visualisation over time - Satellite data overlays (vegetation indices, water surface extent, land use change) - Mobile-friendly layout; exportable as PDF for offline use at Barazas and government meetings

Data sources: pollution episode outputs + monthly sampling outputs + external data (Section 4.7)

Access: - Public view: health scores, pollution incident map, site-level detail, trend data — no login required - Partner view: restricted data layers — SSO or email and password login; no self-registration; accounts created by Admin

Data refresh: weekly

Tech stack: Next.js (frontend), Leaflet.js (maps), ECharts (charts), FastAPI backend API, PostgreSQL database

4.6 Technology Stack

Layer	Technology	Rationale
Data collection — USSD	Africa’s Talking	GSM-only; no smartphone or data required
Data collection — WhatsApp	WhatsApp Business bot (Africa’s Talking)	Menu-driven; same gateway as USSD
Data collection — structured sampling	KoboToolbox	Open source; offline-capable; proven in field contexts

Layer	Technology	Rationale
Satellite data processing	Google Earth Engine	Managed compute; free tier for non-commercial use; no infrastructure to maintain
Database	PostgreSQL + PostGIS	Open source; relational; robust geospatial support
Backend API	FastAPI (Python)	Open source; fast; auto-generates OpenAPI documentation
Frontend — admin and portal	Next.js	Open source; React-based;
Visualisation — charts	ECharts	Open source; performant for time-series and score display
Visualisation — maps	Leaflet.js	Open source; lightweight interactive maps
Infrastructure	Docker	Containerised; portable; supports cloud or in-country hosting
Codebase	GitHub	Version control; issue tracking; CI/CD via GitHub Actions

All components are open-source licensed except the Africa’s Talking API (commercial SaaS) and Google Earth Engine (free tier, non-commercial use).

The platform is designed so that NBD and its national partners can operate and maintain it independently after the pilot period.

4.7 External Data Integration

Purpose: augment citizen-collected ground-truth data with authoritative external datasets.

Known sources — Phase 1:

- **Sentinel 1 & 2** (Earth Observation) — vegetation indices, water surface extent, land use change
- **CHIRPS** — precipitation and climate data
- **Lab QA results** from academic partners — heavy metals, nutrients, biochemical oxygen demand, orthophosphate, nitrate, mercury

Sentinel 1 & 2 imagery and CHIRPS precipitation data are processed using Google Earth Engine (GEE). GEE scripts extract vegetation indices, water surface extent, and land use change indicators and export results as GeoJSON or GeoTIFF for ingestion into the platform database.

Anticipated future sources: - Data from Earthwatch - Additional EO datasets

Integration pattern: - Sentinel 1 & 2 / CHIRPS: automated batch pipeline (scheduled jobs) pulling and processing data via GEE; results ingested as structured records - Lab QA results: submitted by academic partners via the portal webform (Add New → Lab QA report) - Format standards: GeoTIFF (EO data), CSV (lab results), REST API (future sources) - Temporal alignment: monthly

Data provenance: every external dataset record carries source name, collection date, and version. This is displayed alongside derived scores in the portal.

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5. Data Model

5.1 Parameters Collected

Category	Parameters	Collected by
Physico-chemical	pH, Temperature, Dissolved Oxygen	Citizen scientists (handheld multi-parameter probes)
Ecological	Fish Catch Per Unit Effort (CPUE)	Citizen scientists
Hydrological / catchment	Water levels/flow, Flooding extent	Citizen scientists + wetland watchers
Pollution episodes	water colour, smell, fish or other animal kills, or change in water levels	Wetland watchers (USSD/WhatsApp)
Lab validation	Biochemical Oxygen Demand, Orthophosphate, Nitrate, Mercury, Heavy metals and nutrient loads (N/P)	Academic partners (shadow sampling)
Indigenous knowledge	Community observations captured through	CSO staff (via admin FGD session form)

Category	Parameters	Collected by
	FGD sessions at Monthly Barazas	

5.2 Data Schema — Pollution Episode Reporting

[Schema to be defined — fields, types, constraints, units, aligned with Annex 1 of inception report (citizen reporting form)]

5.3 Data Schema — Monthly Structured Water Quality Sampling

[Schema to be defined — aligned with KoboToolbox form design]

5.4 External Data Schema

[Schema per external source type — how Sentinel 1&2 / CHIRPS / lab records join to citizen science records via site identifier and date]

5.5 Site Identifiers

Persistent, structured site identifiers follow the format `NBD-MARA-001`, `NBD-SIO-001`, etc. Four sampling sites per wetland. The site ID must be consistent across all data layers.

The site ID is the primary key for interoperability. It is embedded in every layer of the platform:

```
KoboCollect form      →  site_id field (fixed, pre-loaded per device)
          │
          ▼
Admin layer record    →  site_id: NBD-MARA-001
          │
          ├──▶ GEE pipeline output    {"site_id": "NBD-MARA-001", "ndvi": 
0.72}
          │
          ├──▶ Lab QA record          site_id: NBD-MARA-001, period: 2026-06
          │
          ├──▶ FGD session record     site_id: NBD-MARA-001, period: 2026-06
          │
          └──▶ Portal API response    GET /api/sites/NBD-MARA-001/scores
```

Any external dataset can join to platform data using the site ID alone. No custom field mapping is needed.

5.6 Wetland Health Scores

Health scores are calculated at the parameter level, aggregated into group means, and combined into a single composite score per site.

Health Classes (upper bound inclusive)

Class	Label	Score range	Description
A	Very Good / Natural	0.8 – 1.0	Unmodified or natural; very high ecological integrity
B	Good / Slightly modified	0.6 – 0.8	Largely natural with few modifications; small loss of natural habitat
C	Moderate / Moderately modified	0.4 – 0.6	Moderate change in ecosystem processes and loss of natural habitats
D	Poor / Largely modified	0.2 – 0.4	Large change in ecosystem processes; serious loss of natural habitat and biota
E	Very Poor / Critically modified	0.0 – 0.2	Critical modifications; ecosystem processes completely altered

Parameter Groups and Scoring

Parameter group	Parameters scored
Physico-chemical	Water temperature, pH, Dissolved Oxygen (converted to Water Quality Index score)
Catchment and hydrological	% wetland converted to non-wetland use; r
Ecological	% wetland area covered by invasive macrophytes; Fish Catch Per Unit Effort;

Each group is scored on a 0.0–1.0 scale. Group scores are averaged to produce the composite score.

Scoring Pipeline

```
Monthly sampling record (approved)
           │
           ▼
┌──────────────────────────────────────────────────┐
│  Score each parameter group (0.0–1.0)    │        │
│                                                   │
│  Physico-chemical  ──────────▶ 0.68          │    │
│  Catchment/hydro   ──────────▶ 0.58          │    │
│  Ecological        ──────────▶ 0.65          │    │
│                                          │        │
└──────────────────────────────────────────────────┘
                      │
                      ▼
        Composite = average = 0.638
                      │
     ┌──────────────────────────────────┐
     │  FGD session IK signal     │      │
     │  fish: moderately declined │      │
     │  clarity: much worse       │      │
     │  vegetation: partially lost│      │
     │  encoded signal = 0.67     │      │
     └──────────────────────────────────┘
                      │
                      ▼
      Fuzzy logic adjustment
      0.638 → 0.55
                      │
                      ▼
     ┌────────────────────────────────────┐
     │  Health class: C (Moderate)    │    │
     │  Traffic light: YELLOW        │     │
     │  Action: community response   │     │
     └────────────────────────────────────┘
```

Fuzzy Logic Adjustment

A composite score of 0.638 sits on the border between Yellow and Green. The number alone does not capture whether the wetland is degrading relative to its historical state. Indigenous knowledge from FGD sessions at Monthly Barazas provides that context as a soft signal.

Step 1 — Encode the IK signal.

FGD session records use structured dropdowns per dimension. Each response maps to a numeric score. The scores are averaged into a single IK signal (0.0 = no decline reported; 1.0 = severe decline reported).

Dimension	FGD response	Encoded score
Fish abundance	Moderately declined	0.6
Water clarity	Much worse	1.0
Vegetation cover	Partially lost	0.4
IK signal		**0.67**

Step 2 — Express inputs as fuzzy sets.

Both inputs (composite score and IK signal) are assigned membership values across three fuzzy sets: low, medium, high. A score of 0.638 falls mainly in the "medium" set. An IK signal of 0.67 falls mainly in the "moderate" set.

Step 3 — Apply if-then rules.

Rules fire based on the combination of input fuzzy sets:

Composite score	IK signal	Output
High	None	High
High	Moderate	Medium
High	Strong	Medium
Medium	None	Medium
Medium	**Moderate**	**Low** ← fires
Medium	Strong	Low
Low	Any	Low

Step 4 — Defuzzify.

The fired rule output is converted back to a single number using the centroid method.

Result: composite 0.638 → adjusted score **0.55** → traffic light **YELLOW** → community response triggered.

Without the FGD data, 0.638 would classify as Green. No action would be taken.

Traffic Light Classification

Status	Score threshold
Green	*> 0.6*
Yellow	*0.4 - 0.6*
Red	*0 - 0.4*

Management Actions

Management actions are displayed publicly on the portal alongside the traffic light status for each site. Each action shows a short 3-word label as the primary display, with the full description available on expansion. Actions are defined per wetland by the NBD Secretariat and national partners and updated via the admin interface.

Green — No action required. The wetland is within acceptable health bounds. Monitoring continues at the regular monthly cadence.

Yellow

- Establishment of Silt Traps and Grass Strips: Install vegetative filters along the edges of agricultural plots to catch sediment and nutrient runoff (phosphates/nitrates) before they reach the water.
- Constructed Wetlands for Tertiary Treatment: Encourage small-scale, man-made wetlands at the edge of settlement zones to naturally treat domestic greywater.
- Promotion of Livelihoods: Transition farmers in the buffer zone from high-input crops (like sugarcane or maize) to low-impact activities such as apiculture (beekeeping) or sustainable papyrus harvesting.
- Riparian Re-vegetation: Conduct targeted planting of indigenous species (e.g., Ficus sycomorus or Typha) in degraded patches to maintain the 0.4–0.6 ecological integrity score.
- Establishment of Community Conservation Areas (CCAs): Designate no-catch zones or seasonal closures during fish spawning periods to allow stocks to replenish.
- Introduction of Co-Management Groups: Form Beach Management Units (BMUs) that are empowered to self-regulate gear sizes and monitor illegal fishing, reducing the burden on the central Authority.

Red

- Prompt reporting effluent and/or sewage discharge incidences at the local unit of Ministry of Environment and Water.

- Setting up interceptor STPs to direct sewage to sewage treatment plant/combined effluent treatment plant. A plan including the budget and timeline estimates will be presented in the meeting
- Preventing any further encroachment of the wetland via enacting protected area/buffer zone regulation (e.g. increasing buffer widths) as per the Environment Management & Coordination Act. Any violation should be reported immediately to the County Environment Committee.
- Fishermen should switch to better, lower meshed gillnets to increase the catch per unit effort
- The Environment Management Authority should put together a draft management for Wetland for public review by December 2026

The portal shows the 3-word label on the site health card. Clicking it expands the full action text. All Red actions are publicly visible — no login required. The NBD Secretariat updates action text via the admin interface when escalation steps change.

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6. Integration Design

Integration	Direction	Protocol	Notes
Africa's Talking (USSD)	Inbound	HTTP webhook	Stateful session management required
Africa's Talking (WhatsApp Business bot)	Inbound/outbound	HTTP webhook	Menu-driven flow
KoboToolbox API	Outbound pull	REST/JSON	Scheduled daily sync
Wetland data portal data feed	Internal	FastAPI REST endpoints	Admin layer serves data to portal frontend via documented API

Integration	Direction	Protocol	Notes
Sentinel 1 & 2 ingestion	Inbound(Monthly)	GEE export (GeoTIFF / GeoJSON)	Scheduled batch pipeline
CHIRPS ingestion	Inbound(Monthly)	GEE export (GeoJSON)	Batch, scheduled
Lab QA data ingestion	Inbound(Quarterly	Portal webform (Add New → Lab QA report)	

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7. Security, Privacy & Data Governance

7.1 Citizen Data Privacy

All personal identifiable information is stripped before data is displayed on the public portal. Only aggregated data is displayed for pollution episodes.

7.2 Data Classification

Tier	Examples	Who can access
Public	Aggregated wetland health scores, pollution incident map, trend charts	Anyone
Private	Wetland watcher / citizen scientist PII, phone numbers, individual episode links	Admins only

7.3 Access Control Model

Role	Permissions
Wetland watcher	Submit own pollution episode reports only
Citizen scientist	Submit own sampling records via KoboCollect only
Akvo , NBD	Full platform administration
Public	Read public tier only (portal)

Authentication: - Admin interface: SSO via common identity providers (Google, Microsoft). No passwords managed by the platform. - Portal (partner access): email and password login, SSO. - No self-registration. All accounts are created by an Admin via the invite flow. - Public portal views require no login.

7.4 Data Ownership & Sovereignty

Ownership. NBD and the communities from which data is collected are the data owners. Akvo is the data processor — it operates the platform on behalf of NBD .

Akvo's role. Akvo has no right to use platform data for any purpose beyond operating the platform for this project. Akvo will not share, sell, or analyse the data for its own purposes.

7.5 Infrastructure Security

- Secrets management: no credentials in code or Docker images; use environment secrets or a secrets manager
- Encryption at rest: all PII and observation data
- Encryption in transit: TLS for all services

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8. Deployment Architecture

8.1 Docker Services

The platform is fully Dockerised. Each major component runs as a separate Docker service.

8.2 Hosting

The platform is cloud-hosted. The architecture also supports local in-country server deployment should NBD choose to operate it independently post-pilot.

Production server specification: 8 GB RAM, 4-core CPU, 30 GB storage. Media attachments (photos, documents) are stored in cloud object storage, separate from the VM.

Three environments are maintained: development, staging, and production.

Codebase. Maintained on GitHub under the Akvo organisation. Continuous integration via GitHub Actions runs automated tests on every pull request. Deployments to staging and production are triggered via GitHub Actions.

8.3 Backup & Recovery

- Daily automated backups of all platform data
- Backup retention: 30 days

8.4 Monitoring & Alerting

- Infrastructure monitoring: server uptime, CPU and memory utilisation, disk usage

- Application monitoring: Africa's Talking webhook delivery failures, KoboToolbox sync errors, admin layer processing queue depth, portal availability
- Alerting: email notification to Akvo operations team on critical failures

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9. Testing & Validation Strategy

9.1 Technical Testing

- Unit testing: core business logic (health score calculation, validation rules, traffic light classification, data transformations)
- Integration testing: USSD session flows, WhatsApp Business bot flows, KoboToolbox API sync, Sentinel 1 & 2 and CHIRPS ingestion pipelines
- End-to-end testing: full wetland watcher journey (pollution episode) and full citizen scientist journey (monthly sampling)

9.2 Field Validation

- Field pilot with actual wetland watchers and citizen scientists before full rollout; document failure modes
- Train-the-Trainers (ToT) workshop as a system validation checkpoint — system must be usable by ToT-trained users from day one

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10. Project Timeline

Milestone	Date
Design signoff	22 May 2026
Platform development	25 May – 26 June 2026
Training	End of June 2026
Data collection begins	Early July 2026

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11. Forms

Date source 1: Citizen reporting

- USSD based reporting

  Dial XXX to report a change in water colour, smell, fish or other animal kills, or change in water levels

  Press 1 if the water colour suddenly became darker/murkier

Press 2: If the water is smelling bad

Press 3: If you see many dead fishes or animals in the river

Press 4: If the water is level is too high

Press 5: If the water level is too low after low flows

- Additional information in case of WhatsApp based reporting

  Take at least 2 photos of your observation

  Add voice notes/text comments if you deem necessary (Optional):

Data source 2: Citizen scientist (In Kobo)

Site details:

- Sampling Id
- Photo
- GPS reading
- Water quality
  - pH, Water Temperature and Dissolved Oxygen test results
    - Boundaries : pH= 2-10, Water Temp: 5-50 degree C, Dissolved Oxygen= 0.5- 35 mg/L
- Catchment
  - Type of crops grown in the catchment
  - Type of plant species in the catchment
- Ecological
  - % wetland site covered by invasive macrophytes
  - Catch per unit effort
    - Target species: Mainly, which fish species (X) are you capturing?

        - Total catch: How many (Y) of those species were caught?
        - Effort: How many hours did you spend actively fishing and getting the XY catch? (exclude travel/rest time)

- Hydrological
    - What is the water level during the time of sampling?
        - High (if the riverbanks are not visible and floodplains are inundated)
        - Medium (if the banks are visible and floodplains are not inundated)
        - Low (if the water has trickled down a small stream/narrow river and parts of the river bed are visible)